BISMILLAH SELESAI DALAM 3 JAM

11.
jalankan diutahime

```
cat > /root/utahime-dns.sh << 'EOF'
#!/bin/bash
export DEBIAN_FRONTEND=noninteractive
apt-get update; apt-get install -y bind9 bind9utils bind9-doc dnsutils

# Buat direktori untuk file zona
mkdir -p /etc/bind/jujutsu

# Konfigurasi zona master dan izinkan transfer ke slave (shoko)
cat > /etc/bind/named.conf.local << 'EOT'
zone "jujutsu.K53.com" {
    type master;
    file "/etc/bind/jujutsu/jujutsu.com";
    allow-transfer { 10.90.1.3; };
    notify yes;
};
EOT

# Buat file zona dengan SOA dan NS record
cat > /etc/bind/jujutsu/jujutsu.com << 'EOT'
$TTL    604800
@       IN      SOA     utahime.jujutsu.K53.com. root.jujutsu.K53.com. (
                         2026101401 ; Serial
                            604800 ; Refresh
                             86400 ; Retry
                           2419200 ; Expire
                            604800 ) ; Negative Cache TTL
;
@       IN      NS      utahime.jujutsu.K53.com.
@       IN      NS      shoko.jujutsu.K53.com.
utahime IN      A       10.90.1.2
shoko   IN      A       10.90.1.3
EOT

# Izinkan query dari semua network internal
sed -i 's/listen-on-v6 { any; };/listen-on-v6 { any; };\n\tallow-query { any; };/'
/etc/bind/named.conf.options

service bind9 restart
```

EOF

bash /root/utahime-dns.sh



jalankan di shoko

cat > /root/shoko-dns.sh << 'EOF'
#!/bin/bash
export DEBIAN_FRONTEND=noninteractive
apt-get update; apt-get install -y bind9 bind9utils bind9-doc dnsutils

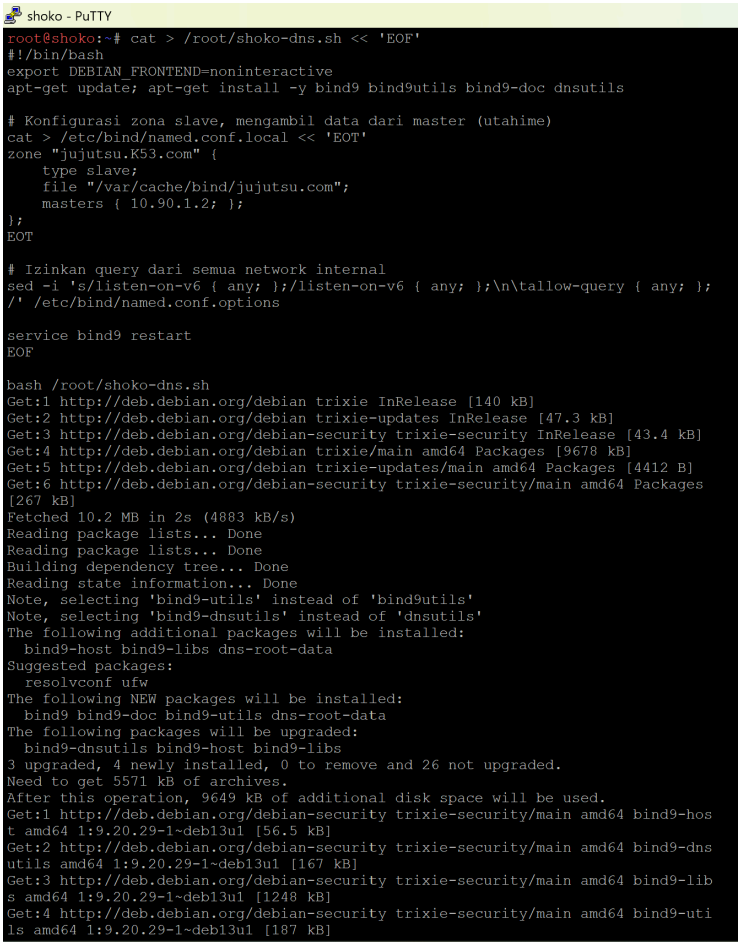
Konfigurasi zona slave, mengambil data dari master (utahime)
cat > /etc/bind/named.conf.local << 'EOT'
zone "jujutsu.K53.com" {
type slave;
file "/var/cache/bind/jujutsu.com";
masters { 10.90.1.2; };
};
EOT

Izinkan query dari semua network internal

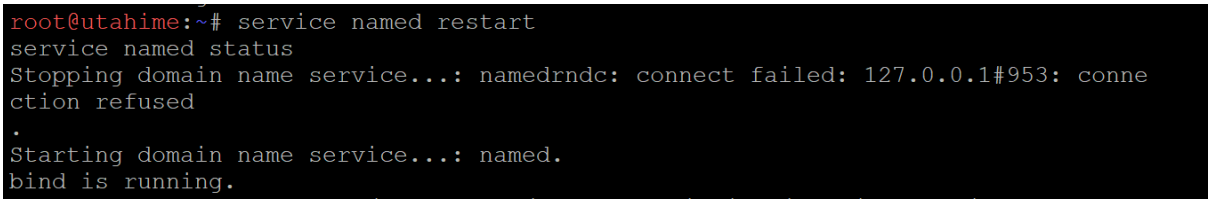
sed -i 's/listen-on-v6 { any; };/listen-on-v6 { any; };\n\tallow-query { any; };/' /etc/bind/named.conf.options

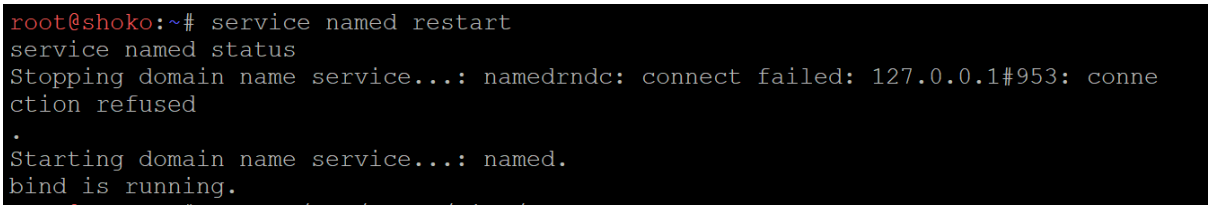
service bind9 restart
EOF

bash /root/shoko-dns.sh



kalau error ulangi diutahime dan shoko
service named restart
service named status





- Verifikasi Sinkronisasi dan Kenaikan Nomor Serial

jalankan dishoko
ls -l /var/cache/bind/
dig @127.0.0.1 jujutsu.K53.com SOA +short

```
root@shoko:~# ls -l /var/cache/bind/
dig @127.0.0.1 jujutsu.K53.com SOA +short
total 8
-rw-r--r-- 1 bind bind  319 Oct 14 12:04 jujutsu.com
-rw-r--r-- 1 bind bind 1336 Oct 14 12:04 managed-keys.bind.jnl
utahime.jujutsu.K53.com. root.jujutsu.K53.com. 2026101401 604800 86400 2419200 6
04800
```

jalankan diutahime
sed -i 's/2026101401/2026101402/' /etc/bind/jujutsu/jujutsu.com
rndc reload

```
root@utahime:~# sed -i 's/2026101401/2026101402/' /etc/bind/jujutsu/jujutsu.com
rndc reload
server reload successful
```

lalu cek lagi dishoko
dig @127.0.0.1 jujutsu.K53.com SOA +short

```
root@shoko:~# dig @127.0.0.1 jujutsu.K53.com SOA +short
utahime.jujutsu.K53.com. root.jujutsu.K53.com. 2026101402 604800 86400 2419200 6
04800
```

- Verifikasi Query DNS dari Client

jalankan diitadori
apk add bind-tools

```
itadori:~# apk add bind-tools
WARNING: updating and opening https://dl-cdn.alpinelinux.org/alpine/v3.23/main/x
86_64/APKINDEX.tar.gz: DNS: transient error (try again later)
WARNING: updating and opening https://dl-cdn.alpinelinux.org/alpine/v3.23/commun
ity/x86_64/APKINDEX.tar.gz: DNS: transient error (try again later)
OK: 81.3 MiB in 165 packages
```

Query ke Master (utahime)
dig @10.90.1.2 jujutsu.K53.com NS +short

```
root@utahime:~# dig @10.90.1.2 jujutsu.K53.com NS +short
shoko.jujutsu.K53.com.
utahime.jujutsu.K53.com.
```

Query ke Slave (shoko)
dig @10.90.1.3 jujutsu.K53.com NS +short

```
root@shoko:~# dig @10.90.1.3 jujutsu.K53.com NS +short
utahime.jujutsu.K53.com.
shoko.jujutsu.K53.com.
```

```
itadori:~# dig @10.90.1.2 jujutsu.K53.com NS +short
shoko.jujutsu.K53.com.
utahime.jujutsu.K53.com.
itadori:~# dig @10.90.1.3 jujutsu.K53.com NS +short
shoko.jujutsu.K53.com.
utahime.jujutsu.K53.com.
```

12.
jalankan diutahime

```
cat > /root/utahime-soal12.sh << 'EOF'
#!/bin/bash

# 1. Update Zona Forward (tambah A record gojo dan CNAME www)
# Serial dinaikkan menjadi 2026101403 agar shoko otomatis update
cat > /etc/bind/jujutsu/jujutsu.com << 'EOT'
$TTL    604800
@       IN      SOA     utahime.jujutsu.K53.com. root.jujutsu.K53.com. (
                         2026101403 ; Serial
                            604800 ; Refresh
                             86400 ; Retry
                           2419200 ; Expire
                            604800 ) ; Negative Cache TTL
;
@       IN      NS      utahime.jujutsu.K53.com.
@       IN      NS      shoko.jujutsu.K53.com.
@       IN      A       10.90.1.4
utahime IN      A       10.90.1.2
shoko   IN      A       10.90.1.3
gojo    IN      A       10.90.1.4
www     IN      CNAME   jujutsu.K53.com.
EOT

# 2. Tambah Zona Reverse ke named.conf.local
cat >> /etc/bind/named.conf.local << 'EOT'

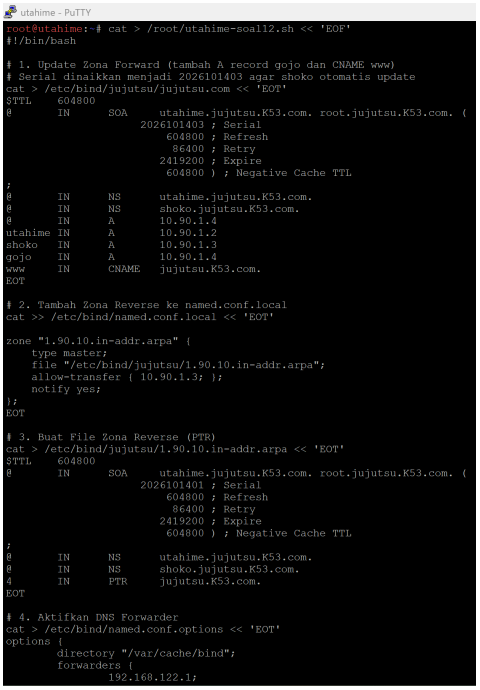
zone "1.90.10.in-addr.arpa" {
    type master;
    file "/etc/bind/jujutsu/1.90.10.in-addr.arpa";
    allow-transfer { 10.90.1.3; };
    notify yes;
};
EOT
```

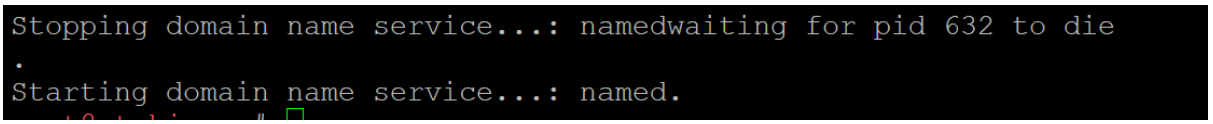
```
# 3. Buat File Zona Reverse (PTR)
cat > /etc/bind/jujutsu/1.90.10.in-addr.arpa << 'EOT'
$TTL    604800
@       IN      SOA     utahime.jujutsu.K53.com. root.jujutsu.K53.com. (
                         2026101401 ; Serial
                            604800 ; Refresh
                             86400 ; Retry
                           2419200 ; Expire
                            604800 ) ; Negative Cache TTL
;
@       IN      NS      utahime.jujutsu.K53.com.
@       IN      NS      shoko.jujutsu.K53.com.
4       IN      PTR     jujutsu.K53.com.
EOT

# 4. Aktifkan DNS Forwarder
cat > /etc/bind/named.conf.options << 'EOT'
options {
        directory "/var/cache/bind";
        forwarders {
                192.168.122.1;
        };
        forward only;
        dnssec-validation auto;
        listen-on-v6 { any; };
        allow-query { any; };
};
EOT

service named restart
EOF

bash /root/utahime-soal12.sh
```





jalankan dishoko

```
cat > /root/shoko-soal12.sh << 'EOF'
#!/bin/bash

# 1. Tambah Zona Reverse sebagai slave
cat >> /etc/bind/named.conf.local << 'EOT'

zone "1.90.10.in-addr.arpa" {
    type slave;
    file "/var/cache/bind/1.90.10.in-addr.arpa";
    masters { 10.90.1.2; };
};
EOT

# 2. Aktifkan DNS Forwarder
cat > /etc/bind/named.conf.options << 'EOT'
options {
        directory "/var/cache/bind";
        forwarders {
                192.168.122.1;
        };
        forward only;
```

```
        dnssec-validation auto;
        listen-on-v6 { any; };
        allow-query { any; };
};
EOT

service named restart
EOF
```

bash /root/shoko-soal12.sh

```
root@shoko:~# cat > /root/shoko-soal12.sh << 'EOF'
#!/bin/bash

# 1. Tambah Zona Reverse sebagai slave
cat >> /etc/bind/named.conf.local << 'EOT'

zone "1.90.10.in-addr.arpa" {
    type slave;
    file "/var/cache/bind/1.90.10.in-addr.arpa";
    masters { 10.90.1.2; };
};
EOT

# 2. Aktifkan DNS Forwarder
cat > /etc/bind/named.conf.options << 'EOT'
options {
        directory "/var/cache/bind";
        forwarders {
                192.168.122.1;
        };
        forward only;
        dnssec-validation auto;
        listen-on-v6 { any; };
        allow-query { any; };
};
EOT

service named restart
EOF

bash /root/shoko-soal12.sh
Stopping domain name service...: namedwaiting for pid 630 to die
.
Starting domain name service...: named.
```

Jalankan diitadori

1. Cek CNAME:

dig @10.90.1.2 www.jujutsu.K53.com +short

2. Cek Reverse DNS (PTR):

dig @10.90.1.2 -x 10.90.1.4 +short

3. Cek DNS Forwarder (Internet): Karena forwarder sudah menyala, klien seharusnya bisa ping ke internet walau hanya menggunakan IP DNS lokal.

ping -c 3 google.com

```
itadori:~# dig @10.90.1.2 www.jujutsu.K53.com +short
jujutsu.K53.com.
10.90.1.4
itadori:~# dig @10.90.1.2 -x 10.90.1.4 +short
jujutsu.K53.com.
itadori:~# ping -c 3 google.com
PING google.com (142.251.175.101) 56(84) bytes of data.
64 bytes from sh-in-f101.1e100.net (142.251.175.101): icmp_seq=1 ttl=100 time=23.2 ms
64 bytes from sh-in-f101.1e100.net (142.251.175.101): icmp_seq=2 ttl=100 time=23.5 ms
64 bytes from sh-in-f101.1e100.net (142.251.175.101): icmp_seq=3 ttl=100 time=23.8 ms

--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2182ms
rtt min/avg/max/mdev = 23.150/23.474/23.760/0.250 ms
```

13.
jalankan diyaga

```
cat > /root/yaga-db.sh << 'EOF'
#!/bin/bash
export DEBIAN_FRONTEND=noninteractive
echo "nameserver 192.168.122.1" > /etc/resolv.conf
apt-get update; apt-get install -y mariadb-server mariadb-client

cat > /etc/mysql/mariadb.conf.d/99-master.cnf << 'EOT'
[mysqld]
bind-address = 0.0.0.0
server-id = 1
log_bin = /var/log/mysql/mysql-bin.log
binlog_do_db = db_shibuya_K53
EOT

mkdir -p /var/log/mysql
chown -R mysql:mysql /var/log/mysql
service mariadb restart || service mysql restart

mysql -u root << 'EOT'
CREATE DATABASE IF NOT EXISTS db_shibuya_K53;
USE db_shibuya_K53;

CREATE TABLE IF NOT EXISTS request_logs (
    id INT AUTO_INCREMENT PRIMARY KEY,
    worker_name VARCHAR(50),
    client_ip VARCHAR(50),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

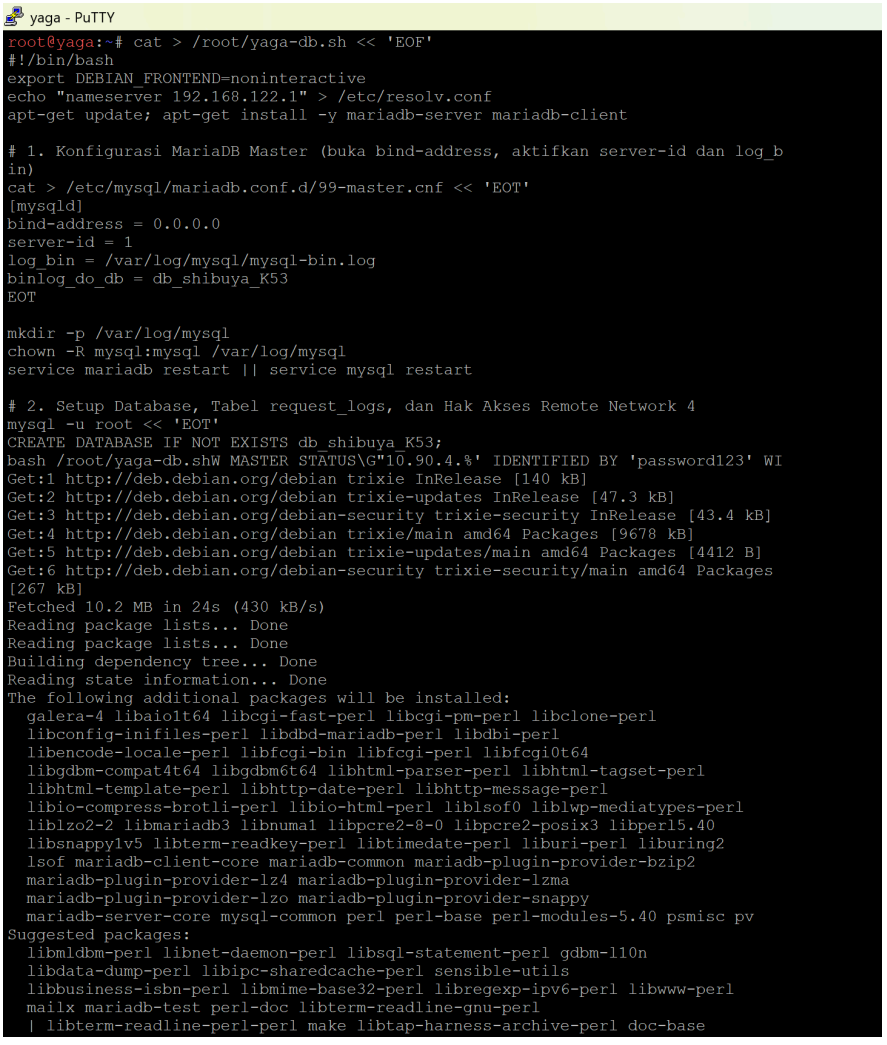
```
CREATE USER IF NOT EXISTS 'repl'@'10.90.4.%' IDENTIFIED BY 'password123';
GRANT REPLICATION SLAVE ON *.* TO 'repl'@'10.90.4.%';

CREATE USER IF NOT EXISTS 'jujutsu'@'10.90.4.%' IDENTIFIED BY
'password123';
GRANT ALL PRIVILEGES ON db_shibuya_K53.* TO 'jujutsu'@'10.90.4.%';

GRANT ALL PRIVILEGES ON *.* TO 'root'@'10.90.4.%' IDENTIFIED BY
'password123' WITH GRANT OPTION;
FLUSH PRIVILEGES;
EOT

echo "=== STATUS MASTER YAGA ==="
mysql -u root -e "SHOW MASTER STATUS\G"
EOF

bash /root/yaga-db.sh
```



jalankan diinumaki

```
cat > /root/inumaki-db.sh << 'EOF'
#!/bin/bash
export DEBIAN_FRONTEND=noninteractive
echo "nameserver 192.168.122.1" > /etc/resolv.conf
apt-get update; apt-get install -y mariadb-server mariadb-client

cat > /etc/mysql/mariadb.conf.d/99-slave.cnf << 'EOT'
[mysqld]
bind-address = 0.0.0.0
server-id = 2
relay-log = /var/log/mysql/mysql-relay-bin.log
log_bin = /var/log/mysql/mysql-bin.log
binlog_do_db = db_shibuya_K53
EOT

mkdir -p /var/log/mysql
chown -R mysql:mysql /var/log/mysql
service mariadb restart || service mysql restart

mysql -u root << 'EOT'
CREATE DATABASE IF NOT EXISTS db_shibuya_K53;
USE db_shibuya_K53;

CREATE TABLE IF NOT EXISTS request_logs (
    id INT AUTO_INCREMENT PRIMARY KEY,
    worker_name VARCHAR(50),
    client_ip VARCHAR(50),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE USER IF NOT EXISTS 'jujutsu'@'10.90.4.%' IDENTIFIED BY
'password123';
GRANT ALL PRIVILEGES ON db_shibuya_K53.* TO 'jujutsu'@'10.90.4.%';
FLUSH PRIVILEGES;
EOT

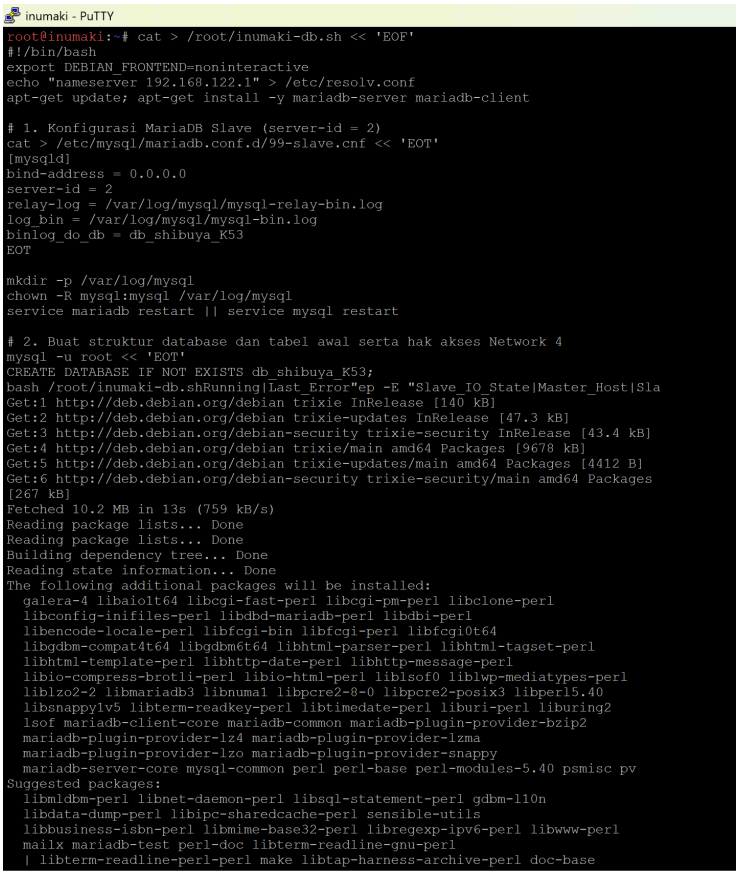
MASTER_STATUS=$(mysql -h 10.90.4.51 -u root -ppassword123 -e "SHOW
MASTER STATUS\G")
LOG_FILE=$(echo "$MASTER_STATUS" | grep "File:" | awk '{print $2}')
LOG_POS=$(echo "$MASTER_STATUS" | grep "Position:" | awk '{print $2}')
```

```
echo "Menghubungkan ke Master dengan File: $LOG_FILE dan Position:
$LOG_POS"

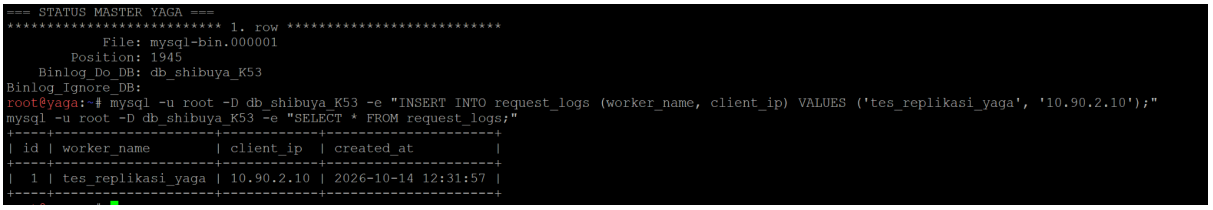
mysql -u root << EOT
STOP SLAVE;
CHANGE MASTER TO
   MASTER_HOST='10.90.4.51',
   MASTER_USER='repl',
   MASTER_PASSWORD='password123',
   MASTER_LOG_FILE='$LOG_FILE',
   MASTER_LOG_POS=$LOG_POS;
START SLAVE;
EOT

sleep 2
echo "=== STATUS SLAVE INUMAKI ==="
mysql -u root -e "SHOW SLAVE STATUS\G" | grep -E
"Slave_IO_State|Master_Host|Slave_IO_Running|Slave_SQL_Running|Last_Error"
EOF

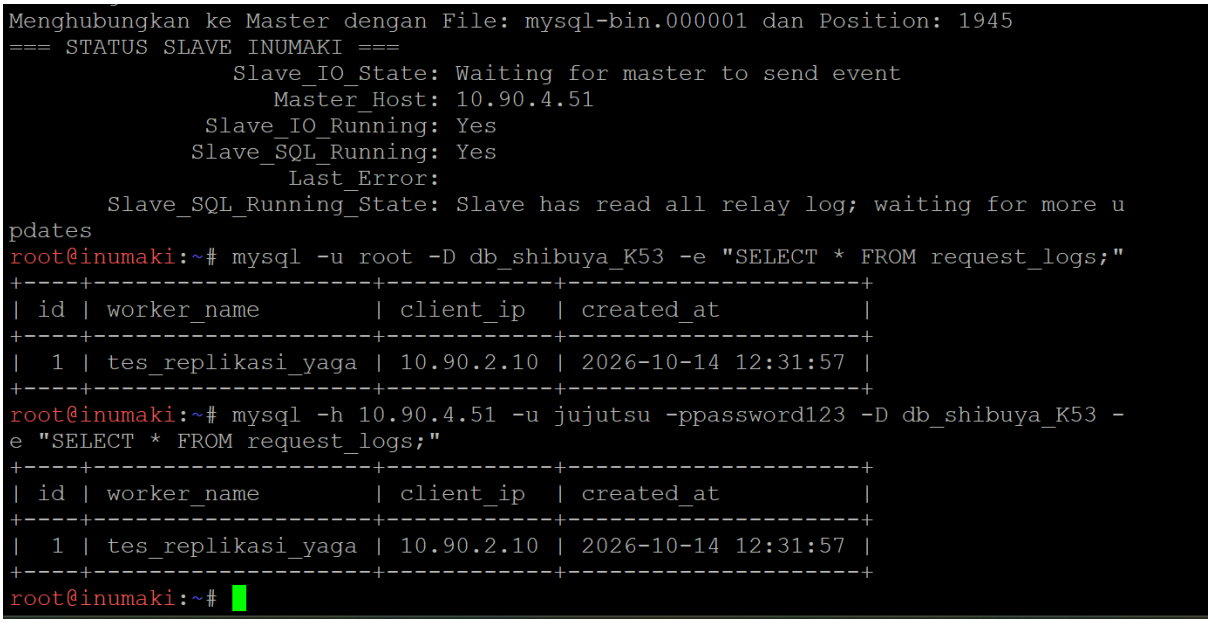
bash /root/inumaki-db.sh
```



verifikasi no 13
jalankan diyaga
mysql -u root -D db_shibuya_K53 -e "INSERT INTO request_logs (worker_name, client_ip) VALUES ('tes_replikasi_yaga', '10.90.2.10');"
mysql -u root -D db_shibuya_K53 -e "SELECT * FROM request_logs;"



jalankan diinumaki
1. Cek apakah data dari yaga sudah otomatis masuk ke inumaki
mysql -u root -D db_shibuya_K53 -e "SELECT * FROM request_logs;"
2. Bukti hak akses remote dari node Network 4 menuju yaga (10.90.4.51)
mysql -h 10.90.4.51 -u jujutsu -ppassword123 -D db_shibuya_K53 -e "SELECT * FROM request_logs;"



14.
jalankan dimaki dan panda

```
cat > /root/web-worker.sh << 'EOF'
#!/bin/bash
export DEBIAN_FRONTEND=noninteractive
echo "nameserver 192.168.122.1" > /etc/resolv.conf
apt-get update; apt-get install -y nginx php-fpm curl

mkdir -p /var/www/jujutsu /run/php

cat > /var/www/jujutsu/index.php << 'EOT'
<?php
header('Content-Type: text/plain');
echo "Jujutsu High Web Frontend - Dilayani oleh: " . gethostname() . "\n";
?>
EOT

service php8.4-fpm start || /etc/init.d/php8.4-fpm start

PHP_SOCK=$(find /run/php/ -name "php*-fpm.sock" | head -n 1)
[ -z "$PHP_SOCK" ] && PHP_SOCK="/run/php/php8.4-fpm.sock"

cat > /etc/nginx/sites-available/jujutsu << EOT
server {
    listen 80;
    server_name jujutsu.K53.com www.jujutsu.K53.com _;

    root /var/www/jujutsu;
    index index.php index.html;

    allow 10.90.1.4;
    allow 127.0.0.1;
    deny all;

    location / {
        try_files \$uri \$uri/ /index.php?\$query_string;
    }

    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:$PHP_SOCK;
    }
```

}
EOT

ln -sf /etc/nginx/sites-available/jujutsu /etc/nginx/sites-enabled/jujutsu
rm -f /etc/nginx/sites-enabled/default

service php8.4-fpm restart
nginx -t && service nginx restart
EOF

bash /root/web-worker.sh

```
root@maki:~# cat > /root/web-worker.sh << 'EOF'
#!/bin/bash
export DEBIAN_FRONTEND=noninteractive
echo "nameserver 192.168.122.1" > /etc/resolv.conf
apt-get update; apt-get install -y nginx php-fpm curl

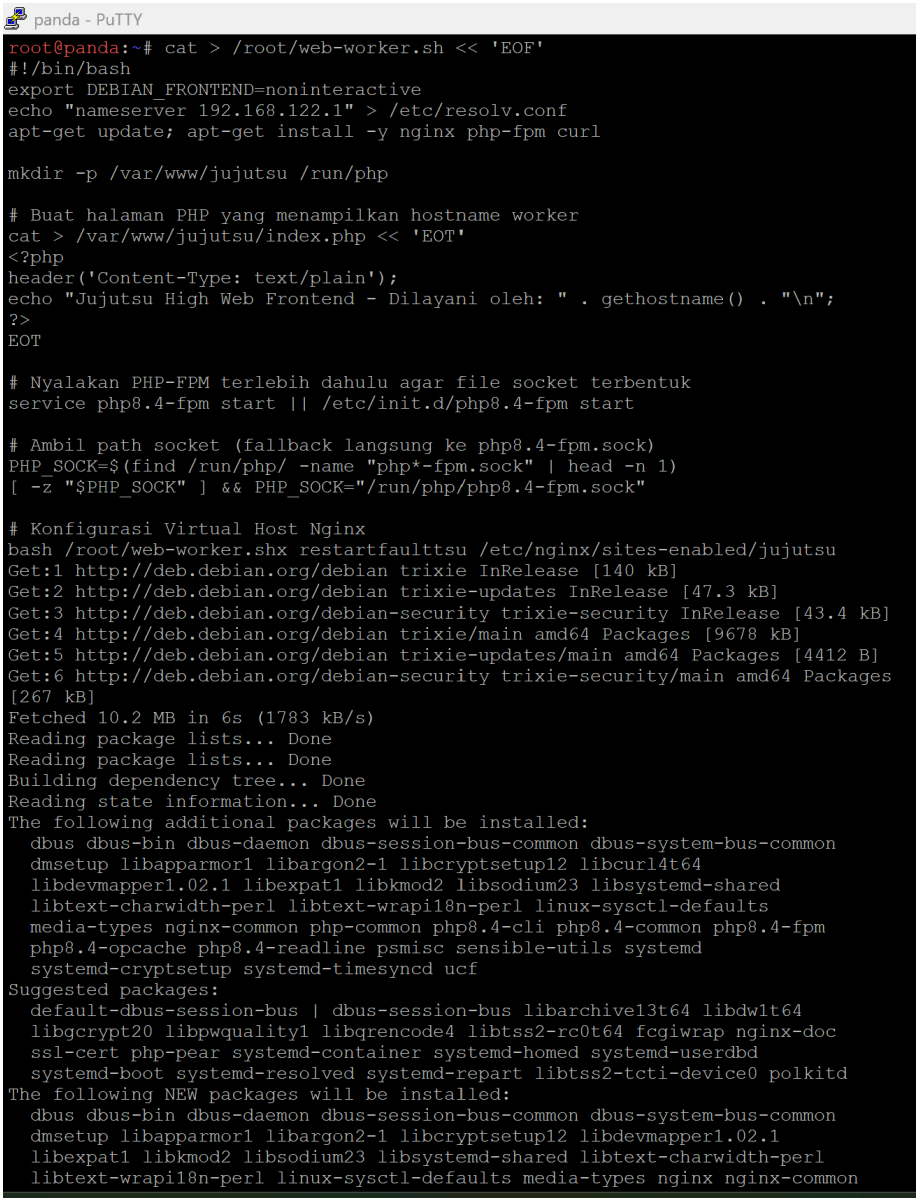
mkdir -p /var/www/jujutsu /run/php

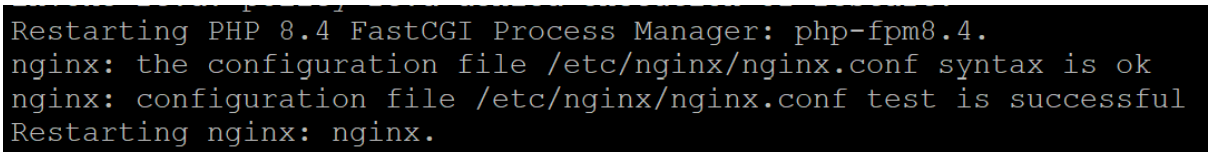
# Buat halaman PHP yang menampilkan hostname worker
cat > /var/www/jujutsu/index.php << 'EOT'
<?php
header('Content-Type: text/plain');
echo "Jujutsu High Web Frontend - Dilayani oleh: " . gethostname() . "\n";
?>
EOT

# Nyalakan PHP-FPM terlebih dahulu agar file socket terbentuk
service php8.4-fpm start || /etc/init.d/php8.4-fpm start

# Ambil path socket (fallback langsung ke php8.4-fpm.sock)
PHP_SOCK=$(find /run/php/ -name "php*-fpm.sock" | head -n 1)
[ -z "$PHP_SOCK" ] && PHP_SOCK="/run/php/php8.4-fpm.sock"

# Konfigurasi Virtual Host Nginx
bash /root/web-worker.shx restartfaulttsu /etc/nginx/sites-enabled/jujutsu
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.26.3-3+deb13u9).
php-fpm is already the newest version (2:8.4+96).
curl is already the newest version (8.14.1-2+deb13u7).
0 upgraded, 0 newly installed, 0 to remove and 27 not upgraded.
Restarting PHP 8.4 FastCGI Process Manager: php-fpm8.4.
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
Restarting nginx: nginx.
```





jalankan dinanami, todo dan kirara

cat > /root/api-worker.sh << 'EOF'
#!/bin/bash
export DEBIAN_FRONTEND=noninteractive
echo "nameserver 192.168.122.1" > /etc/resolv.conf
apt-get update; apt-get install -y nginx php-fpm php-mysql curl

mkdir -p /var/www/api /run/php

```
cat > /var/www/api/index.php << 'EOT'
<?php
header('Content-Type: application/json');

$worker = gethostname();

$client_ip = $_SERVER['REMOTE_ADDR'];
if (!empty($_SERVER['HTTP_X_REAL_IP'])) {
    $client_ip = $_SERVER['HTTP_X_REAL_IP'];
} elseif (!empty($_SERVER['HTTP_X_FORWARDED_FOR'])) {
    $ips = explode(',', $_SERVER['HTTP_X_FORWARDED_FOR']);
    $client_ip = trim($ips[0]);
}

$db_status = "disconnected";
$mysqli = @new mysqli("10.90.4.51", "jujutsu", "password123", "db_shibuya_K53");

if (!$mysqli->connect_error) {
    $stmt = $mysqli->prepare("INSERT INTO request_logs (worker_name, client_ip)
VALUES (?, ?)");
    if ($stmt) {
        $stmt->bind_param("ss", $worker, $client_ip);
        $stmt->execute();
        $stmt->close();
        $db_status = "logged_to_yaga";
    }
    $mysqli->close();
} else {
    $db_status = "error: " . $mysqli->connect_error;
}

echo json_encode([
    "status" => "success",
    "worker" => $worker,
    "client_ip" => $client_ip,
    "db_log" => $db_status
]) . "\n";
?>
EOT

service php8.4-fpm start || /etc/init.d/php8.4-fpm start

PHP_SOCK=$(find /run/php/ -name "php*-fpm.sock" | head -n 1)
[ -z "$PHP_SOCK" ] && PHP_SOCK="/run/php/php8.4-fpm.sock"
```

```
cat > /etc/nginx/sites-available/api << EOT
server {
    listen 80;
    server_name jujutsu.K53.com www.jujutsu.K53.com _;

    root /var/www/api;
    index index.php;

    allow 10.90.4.20;
    allow 127.0.0.1;
    deny all;

    location / {
        try_files \$uri \$uri/ /index.php?\$query_string;
    }

    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:$PHP_SOCK;
    }
}
EOT

ln -sf /etc/nginx/sites-available/api /etc/nginx/sites-enabled/api
rm -f /etc/nginx/sites-enabled/default

service php8.4-fpm restart
nginx -t && service nginx restart
EOF

bash /root/api-worker.sh
```

nanami - PuTTY

```
if (!empty($_SERVER['HTTP_X_REAL_IP'])) {
    $client_ip = $_SERVER['HTTP_X_REAL_IP'];
} elseif (!empty($_SERVER['HTTP_X_FORWARDED_FOR'])) {
    $ips = explode(',', $_SERVER['HTTP_X_FORWARDED_FOR']);
    $client_ip = trim($ips[0]);
}

$db_status = "disconnected";
$mysqli = @new mysqli("10.90.4.51", "jujutsu", "password123", "db_shibuya_K53");

if (!$mysqli->connect_error) {
    $stmt = $mysqli->prepare("INSERT INTO request_logs (worker_name, client_ip) VALUES (?, ?)");
    if ($stmt) {
        $stmt->bind_param("ss", $worker, $client_ip);
        $stmt->execute();
        $stmt->close();
        $db_status = "logged_to_yaga";
    }
    $mysqli->close();
} else {
    $db_status = "error: " . $mysqli->connect_error;
}

echo json_encode([
    "status" => "success",
    "worker" => $worker,
    "client_ip" => $client_ip,
    "db_log" => $db_status
]) . "\n";
?>
EOT

# Nyalakan PHP-FPM terlebih dahulu agar socket terbentuk
service php8.4-fpm start || /etc/init.d/php8.4-fpm start

PHP_SOCK=$(find /run/php/ -name "php*-fpm.sock" | head -n 1)
[ -z "$PHP_SOCK" ] && PHP_SOCK="/run/php/php8.4-fpm.sock"

bash /root/api-worker.shx restartfault/etc/nginx/sites-enabled/api(10.90.4.20)
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.26.3-3+deb13u9).
php-fpm is already the newest version (2:8.4+96).
php-mysql is already the newest version (2:8.4+96).
curl is already the newest version (8.14.1-2+deb13u7).
0 upgraded, 0 newly installed, 0 to remove and 27 not upgraded.
Restarting PHP 8.4 FastCGI Process Manager: php-fpm8.4.
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
Restarting nginx: nginx.
```

todo - PuTTY

```
Creating config file /etc/php/8.4/mods-available/phar.ini with new version
Creating config file /etc/php/8.4/mods-available/posix.ini with new version
Creating config file /etc/php/8.4/mods-available/shmop.ini with new version
Creating config file /etc/php/8.4/mods-available/sockets.ini with new version
Creating config file /etc/php/8.4/mods-available/sysvmsg.ini with new version
Creating config file /etc/php/8.4/mods-available/sysvsem.ini with new version
Creating config file /etc/php/8.4/mods-available/sysvshm.ini with new version
Creating config file /etc/php/8.4/mods-available/tokenizer.ini with new version
Setting up php8.4-opcache (8.4.26-1~deb13u1) ...
Creating config file /etc/php/8.4/mods-available/opcache.ini with new version
Setting up php8.4-readline (8.4.26-1~deb13u1) ...
Creating config file /etc/php/8.4/mods-available/readline.ini with new version
Setting up php8.4-mysql (8.4.26-1~deb13u1) ...
Creating config file /etc/php/8.4/mods-available/mysqlnd.ini with new version
Creating config file /etc/php/8.4/mods-available/mysqli.ini with new version
Creating config file /etc/php/8.4/mods-available/pdo_mysql.ini with new version
Setting up php8.4-cli (8.4.26-1~deb13u1) ...
update-alternatives: using /usr/bin/php8.4 to provide /usr/bin/php (php) in auto
 mode
update-alternatives: warning: skip creation of /usr/share/man/man1/php.1.gz beca
use associated file /usr/share/man/man1/php8.4.1.gz (of link group php) doesn't
exist
update-alternatives: using /usr/bin/phar8.4 to provide /usr/bin/phar (phar) in a
uto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/phar.1.gz bec
ause associated file /usr/share/man/man1/phar8.4.1.gz (of link group phar) doesn
't exist
update-alternatives: using /usr/bin/phar.phar8.4 to provide /usr/bin/phar.phar (
phar.phar) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/phar.phar.1.g
z because associated file /usr/share/man/man1/phar.phar8.4.1.gz (of link group p
har.phar) doesn't exist
Creating config file /etc/php/8.4/cli/php.ini with new version
Setting up php-mysql (2:8.4+96) ...
Setting up php8.4-fpm (8.4.26-1~deb13u1) ...
Creating config file /etc/php/8.4/fpm/php.ini with new version
Created symlink '/etc/systemd/system/multi-user.target.wants/php8.4-fpm.service'
 → '/usr/lib/systemd/system/php8.4-fpm.service'.
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of start.
Setting up php-fpm (2:8.4+96) ...
Setting up libdevmapper1.02.1:amd64 (2:1.02.205-2) ...
Setting up dmsetup (2:1.02.205-2) ...
Setting up libcryptsetup12:amd64 (2:2.7.5-2) ...
Setting up systemd-cryptsetup (257.13-1~deb13u1) ...
Processing triggers for procps (2:4.0.4-9) ...
Processing triggers for libc-bin (2.41-12+deb13u3) ...
Processing triggers for php8.4-cli (8.4.26-1~deb13u1) ...
Processing triggers for php8.4-fpm (8.4.26-1~deb13u1) ...
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of restart.
Restarting PHP 8.4 FastCGI Process Manager: php-fpm8.4.
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
Restarting nginx: nginx.
```

kirara - PuTTY

```
Creating config file /etc/php/8.4/mods-available/phar.ini with new version
Creating config file /etc/php/8.4/mods-available/posix.ini with new version
Creating config file /etc/php/8.4/mods-available/shmop.ini with new version
Creating config file /etc/php/8.4/mods-available/sockets.ini with new version
Creating config file /etc/php/8.4/mods-available/sysvmsg.ini with new version
Creating config file /etc/php/8.4/mods-available/sysvsem.ini with new version
Creating config file /etc/php/8.4/mods-available/sysvshm.ini with new version
Creating config file /etc/php/8.4/mods-available/tokenizer.ini with new version
Setting up php8.4-opcache (8.4.26-1~deb13u1) ...
Creating config file /etc/php/8.4/mods-available/opcache.ini with new version
Setting up php8.4-readline (8.4.26-1~deb13u1) ...
Creating config file /etc/php/8.4/mods-available/readline.ini with new version
Setting up php8.4-mysql (8.4.26-1~deb13u1) ...
Creating config file /etc/php/8.4/mods-available/mysqlnd.ini with new version
Creating config file /etc/php/8.4/mods-available/mysqli.ini with new version
Creating config file /etc/php/8.4/mods-available/pdo_mysql.ini with new version
Setting up php8.4-cli (8.4.26-1~deb13u1) ...
update-alternatives: using /usr/bin/php8.4 to provide /usr/bin/php (php) in auto
 mode
update-alternatives: warning: skip creation of /usr/share/man/man1/php.1.gz beca
use associated file /usr/share/man/man1/php8.4.1.gz (of link group php) doesn't
exist
update-alternatives: using /usr/bin/phar8.4 to provide /usr/bin/phar (phar) in a
uto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/phar.1.gz bec
ause associated file /usr/share/man/man1/phar8.4.1.gz (of link group phar) doesn
't exist
update-alternatives: using /usr/bin/phar.phar8.4 to provide /usr/bin/phar.phar (
phar.phar) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/phar.phar.1.g
z because associated file /usr/share/man/man1/phar.phar8.4.1.gz (of link group p
har.phar) doesn't exist
Creating config file /etc/php/8.4/cli/php.ini with new version
Setting up php-mysql (2:8.4+96) ...
Setting up php8.4-fpm (8.4.26-1~deb13u1) ...
Creating config file /etc/php/8.4/fpm/php.ini with new version
Created symlink '/etc/systemd/system/multi-user.target.wants/php8.4-fpm.service'
 → '/usr/lib/systemd/system/php8.4-fpm.service'.
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of start.
Setting up php-fpm (2:8.4+96) ...
Setting up libdevmapper1.02.1:amd64 (2:1.02.205-2) ...
Setting up dmsetup (2:1.02.205-2) ...
Setting up libcryptsetup12:amd64 (2:2.7.5-2) ...
Setting up systemd-cryptsetup (257.13-1~deb13u1) ...
Processing triggers for procps (2:4.0.4-9) ...
Processing triggers for libc-bin (2.41-12+deb13u3) ...
Processing triggers for php8.4-cli (8.4.26-1~deb13u1) ...
Processing triggers for php8.4-fpm (8.4.26-1~deb13u1) ...
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of restart.
Restarting PHP 8.4 FastCGI Process Manager: php-fpm8.4.
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
Restarting nginx: nginx.
```

- Uji Lokal di Masing-Masing Worker

jalankan dimaki dan panda
curl -s http://127.0.0.1

```
root@maki:~# curl -s http://127.0.0.1
Jujutsu High Web Frontend - Dilayani oleh: maki
```

```
root@panda:~# curl -s http://127.0.0.1
Jujutsu High Web Frontend - Dilayani oleh: panda
```

jalankan dinanami, todo, kirara
curl -s http://127.0.0.1

```
root@nanami:~# curl -s http://127.0.0.1
{"status":"success","worker":"nanami","client_ip":"127.0.0.1","db_log":"logged_t
o_yaga"}
```

```
root@todo:~# curl -s http://127.0.0.1
{"status":"success","worker":"todo","client_ip":"127.0.0.1","db_log":"logged_to_
yaga"}
```

```
root@kirara:~# curl -s http://127.0.0.1
{"status":"success","worker":"kirara","client_ip":"127.0.0.1","db_log":"logged_t
o_yaga"}
```

- Uji Akses Langsung dari Klien Ditolak (403 Forbidden)

jalankan diitadori
apk add curl 2>/dev/null
curl -i http://10.90.4.31
curl -i http://10.90.4.41

kalau diitadori error
killall udhcpc 2>/dev/null
ip addr flush dev eth0
ip link set eth0 up
udhcpc -i eth0

```
itadori:~# apk add curl 2>/dev/null
curl -i http://10.90.4.31
curl -i http://10.90.4.41
OK: 81.3 MiB in 165 packages
HTTP/1.1 403 Forbidden
Server: nginx
Date: Wed, 14 Oct 2026 12:52:29 GMT
Content-Type: text/html
Content-Length: 146
Connection: keep-alive

<html>
<head><title>403 Forbidden</title></head>
<body>
<center><h1>403 Forbidden</h1></center>
<hr><center>nginx</center>
</body>
</html>
HTTP/1.1 403 Forbidden
Server: nginx
Date: Wed, 14 Oct 2026 12:52:29 GMT
Content-Type: text/html
Content-Length: 146
Connection: keep-alive

<html>
<head><title>403 Forbidden</title></head>
<body>
<center><h1>403 Forbidden</h1></center>
<hr><center>nginx</center>
</body>
</html>
```

- Uji Akses Lewat Jalur Proxy Resmi Diizinkan (200 OK)

jalankan digojo

apt-get update; apt-get install -y curl

curl -i http://10.90.4.31

curl -i http://10.90.4.32

```
root@gojo:~# curl -i http://10.90.4.31
curl -i http://10.90.4.32
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 14 Oct 2026 12:54:38 GMT
Content-Type: text/plain;charset=UTF-8
Transfer-Encoding: chunked
Connection: keep-alive

Jujutsu High Web Frontend - Dilayani oleh: maki
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 14 Oct 2026 12:54:38 GMT
Content-Type: text/plain;charset=UTF-8
Transfer-Encoding: chunked
Connection: keep-alive

Jujutsu High Web Frontend - Dilayani oleh: panda
```

jalankan digeto

apt-get update; apt-get install -y curl

curl -i http://10.90.4.41

curl -i http://10.90.4.42

curl -i http://10.90.4.43

```
geto - PuTTY
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages
[267 kB]
Fetched 10.2 MB in 2s (5947 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcurl4t64
The following packages will be upgraded:
  curl libcurl4t64
2 upgraded, 0 newly installed, 0 to remove and 27 not upgraded.
Need to get 662 kB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 http://deb.debian.org/debian-security trixie-security/main amd64 curl amd6
4 8.14.1-2+deb13u7 [270 kB]
Get:2 http://deb.debian.org/debian-security trixie-security/main amd64 libcurl4t
64 amd64 8.14.1-2+deb13u7 [392 kB]
Fetched 662 kB in 0s (3920 kB/s)
(Reading database ... 10187 files and directories currently installed.)
Preparing to unpack .../curl_8.14.1-2+deb13u7_amd64.deb ...
Unpacking curl (8.14.1-2+deb13u7) over (8.14.1-2+deb13u4) ...
Preparing to unpack .../libcurl4t64_8.14.1-2+deb13u7_amd64.deb ...
Unpacking libcurl4t64:amd64 (8.14.1-2+deb13u7) over (8.14.1-2+deb13u4) ...
Setting up libcurl4t64:amd64 (8.14.1-2+deb13u7) ...
Setting up curl (8.14.1-2+deb13u7) ...
Processing triggers for libc-bin (2.41-12+deb13u3) ...
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 14 Oct 2026 12:55:01 GMT
Content-Type: application/json
Transfer-Encoding: chunked
Connection: keep-alive

{"status":"success","worker":"nanami","client_ip":"10.90.4.20","db_log":"logged_
to_yaga"}
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 14 Oct 2026 12:55:01 GMT
Content-Type: application/json
Transfer-Encoding: chunked
Connection: keep-alive

{"status":"success","worker":"todo","client_ip":"10.90.4.20","db_log":"logged_to
_yaga"}
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 14 Oct 2026 12:55:01 GMT
Content-Type: application/json
Transfer-Encoding: chunked
Connection: keep-alive

{"status":"success","worker":"kirara","client_ip":"10.90.4.20","db_log":"logged_
to_yaga"}
```

15.
jalankan digeto

```
cat > /root/geto-lb.sh << 'EOF'
#!/bin/bash
export DEBIAN_FRONTEND=noninteractive
echo "nameserver 192.168.122.1" > /etc/resolv.conf
apt-get update; apt-get install -y nginx curl

# Konfigurasi Upstream API Cluster dengan Weighted Round Robin (nanami
weight=3, todo weight=1)
cat > /etc/nginx/sites-available/load-balancer << 'EOT'
upstream api_cluster {
    server 10.90.4.41 weight=3;
    server 10.90.4.42 weight=1;
}

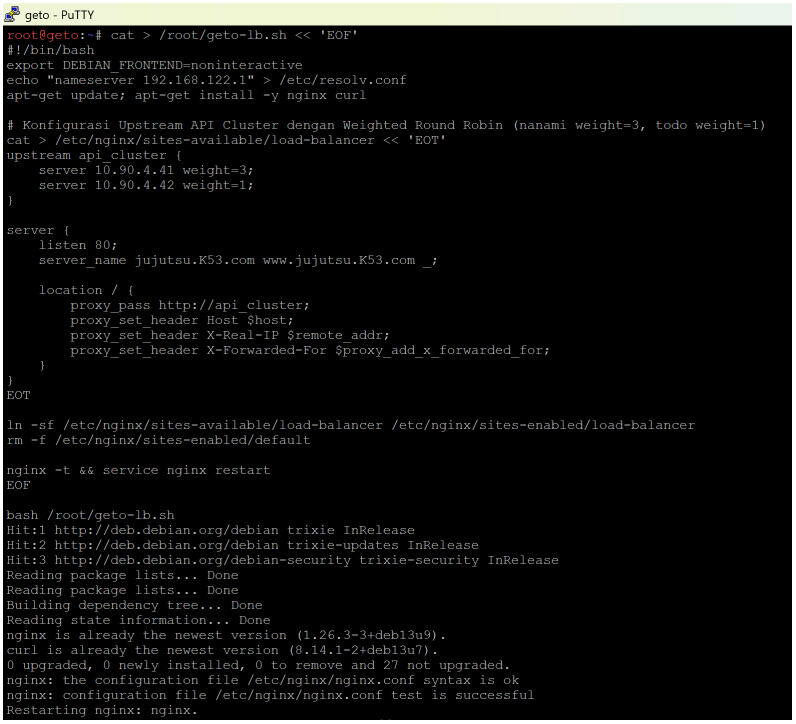
server {
    listen 80;
    server_name jujutsu.K53.com www.jujutsu.K53.com _;

    location / {
        proxy_pass http://api_cluster;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
}
EOT

ln -sf /etc/nginx/sites-available/load-balancer /etc/nginx/sites-enabled/load-balancer
rm -f /etc/nginx/sites-enabled/default

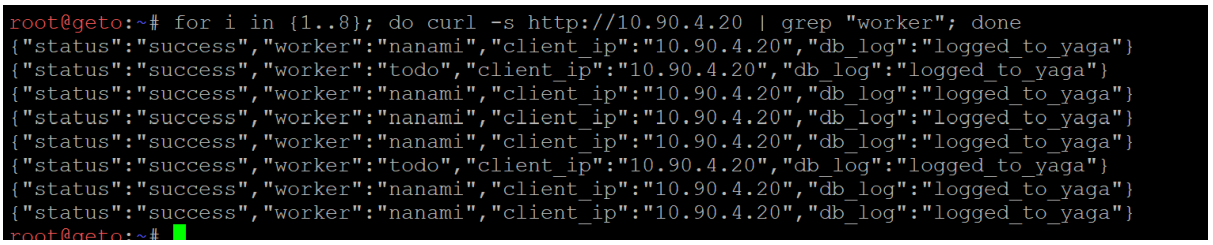
nginx -t && service nginx restart
EOF

bash /root/geto-lb.sh
```



cek digeto juga

for i in {1..8}; do curl -s http://10.90.4.20 | grep "worker"; done



16.

jalankan di gojo

```
cat > /root/gojo-proxy.sh << 'EOF'
#!/bin/bash
export DEBIAN_FRONTEND=noninteractive
echo "nameserver 192.168.122.1" > /etc/resolv.conf
apt-get update; apt-get install -y nginx curl

# Konfigurasi Reverse Proxy & Load Balancer Utama pada gojo
cat > /etc/nginx/sites-available/gojo-proxy << 'EOT'
upstream web_cluster {
    server 10.90.4.31;
    server 10.90.4.32;
}

server {
    listen 80;
    server_name jujutsu.K53.com www.jujutsu.K53.com _;

    # Arahkan trafik berawalan /api/ ke Internal Load Balancer geto (10.90.4.20)
    location /api/ {
        proxy_pass http://10.90.4.20/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }

    # Arahkan trafik web utama ke cluster web frontend (maki & panda)
    location / {
        proxy_pass http://web_cluster;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
}
EOT

ln -sf /etc/nginx/sites-available/gojo-proxy /etc/nginx/sites-enabled/gojo-proxy
rm -f /etc/nginx/sites-enabled/default
```

nginx -t && service nginx restart
EOF

bash /root/gojo-proxy.sh



Cara Verifikasi / Pembuktian Soal 16
jalankan di itadori

1. Uji Akses Web Frontend lewat gojo (10.90.1.4):

curl -i http://10.90.1.4/

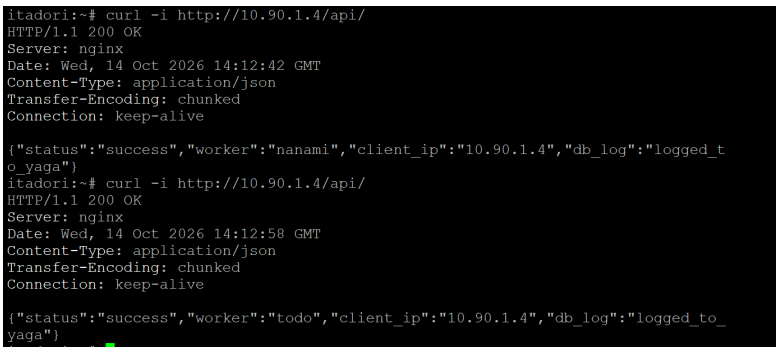
```
itadori:~# curl -i http://10.90.1.4/
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 14 Oct 2026 14:12:16 GMT
Content-Type: text/plain;charset=UTF-8
Transfer-Encoding: chunked
Connection: keep-alive

Jujutsu High Web Frontend - Dilayani oleh: maki
itadori:~# curl -i http://10.90.1.4/
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 14 Oct 2026 14:12:18 GMT
Content-Type: text/plain;charset=UTF-8
Transfer-Encoding: chunked
Connection: keep-alive

Jujutsu High Web Frontend - Dilayani oleh: panda
```

2. Uji Akses API lewat gojo (yang diteruskan ke geto lalu ke nanami/todo):

curl -i http://10.90.1.4/api/



17.

jalankan dinanami, todo, kirara
cat > /root/api-realip.sh << 'EOF'
#!/bin/bash
cat > /var/www/api/index.php << 'EOT'
<?php
header('Content-Type: application/json');
$worker = gethostname();

// Hanya percaya X-Real-IP yang dipasang geto dari proxy terpercaya
$client_ip = $_SERVER['HTTP_X_REAL_IP'] ?? $_SERVER['REMOTE_ADDR'];
if (!filter_var($client_ip, FILTER_VALIDATE_IP)) {
    $client_ip = $_SERVER['REMOTE_ADDR'];
}

$db_status = "disconnected";
$mysqli = @new mysqli("10.90.4.51", "jujutsu", "password123", "db_shibuya_K53");
if (!$mysqli->connect_error) {
    $stmt = $mysqli->prepare("INSERT INTO request_logs (worker_name, client_ip) VALUES (?, ?)");
    if ($stmt) {
        $stmt->bind_param("ss", $worker, $client_ip);
        $stmt->execute();
        $stmt->close();
        $db_status = "logged_to_yaga";
    }
    $mysqli->close();
} else {
    $db_status = "error: " . $mysqli->connect_error;
}

echo json_encode([
    "status" => "success",
    "worker" => $worker,

```
    "client_ip" => $client_ip,
    "db_log" => $db_status
]) . "\n";
EOT

service php8.4-fpm restart
EOF

bash /root/api-realip.sh
```

jalankan digeto

```
cat > /root/geto-lb.sh << 'EOF'
#!/bin/bash
export DEBIAN_FRONTEND=noninteractive
echo "nameserver 192.168.122.1" > /etc/resolv.conf
apt-get update; apt-get install -y nginx curl

cat > /etc/nginx/sites-available/load-balancer << 'EOT'
upstream api_cluster {
    server 10.90.4.41 weight=3 max_fails=2 fail_timeout=5s;
    server 10.90.4.42 weight=1 max_fails=2 fail_timeout=5s;
    server 10.90.4.43 backup;
}

server {
    listen 80;
    server_name jujutsu.K53.com www.jujutsu.K53.com _;

    # Hanya gojo (10.90.1.4) yang dipercaya untuk memberi IP client asli
    set_real_ip_from 10.90.1.4;
    real_ip_header X-Real-IP;

    location / {
        proxy_pass http://api_cluster;
        proxy_next_upstream error timeout http_502 http_503 http_504;
        proxy_next_upstream_tries 3;
        proxy_connect_timeout 2s;

        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
```

```
    }
}
EOT

ln -sf /etc/nginx/sites-available/load-balancer /etc/nginx/sites-enabled/load-balancer
rm -f /etc/nginx/sites-enabled/default
nginx -t && service nginx restart
EOF

bash /root/geto-lb.sh
```

Langkah 3: Verifikasi Pengujian dari Klien (itadori & kusakabe)

1. Uji dari Network 2 (itadori):
2. Bash

```
curl -s http://jujutsu.K53.com/api/
```

3.
4. Harus mencatat client_ip sesuai IP itadori (contoh: 10.90.2.10).
5. Uji dari Network 3 (kusakabe atau miwa): (Jika belum mendapat IP, jalankan udhcpc -i eth0 terlebih dahulu)
6. Bash

```
udhcpc -i eth0

curl -s http://jujutsu.K53.com/api/
```

7.
8. Harus mencatat client_ip sesuai IP Divisi B (contoh: 10.90.3.10 atau 10.90.3.99).
9. Uji Spoofing (Keamanan Header): Kirim header palsu secara sengaja dari itadori untuk membuktikan bahwa geto menolaknya dan tetap meneruskan IP asli yang benar:
10. Bash

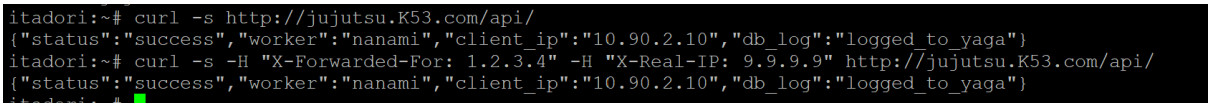
```
curl -s -H "X-Forwarded-For: 1.2.3.4" -H "X-Real-IP: 9.9.9.9"
http://jujutsu.K53.com/api/
```

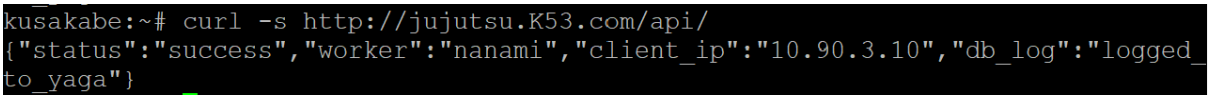
11.

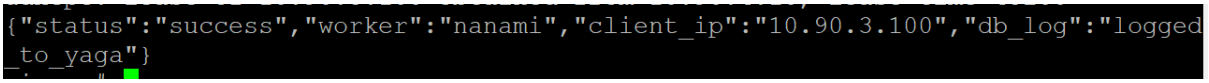
12. Hasil client_ip di JSON harus tetap IP asli itadori, bukan 1.2.3.4 atau 9.9.9.9.
13. Cek di Database (yaga & inumaki): Buka konsol yaga atau inumaki, lalu jalankan query:
14. Bash

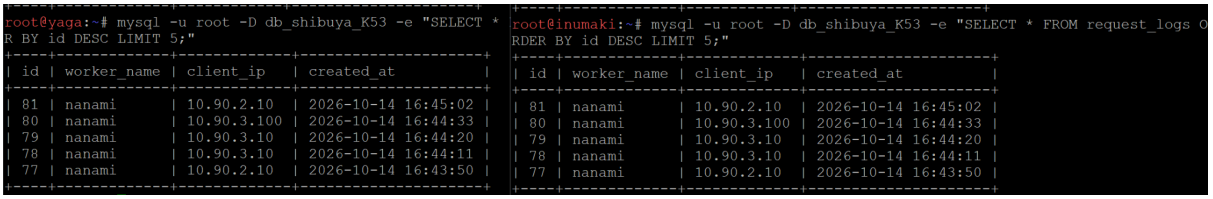
mysql -u root -D db_shibuya_K53 -e "SELECT * FROM request_logs ORDER BY id DESC LIMIT 5;"

15.
16. Pastikan kolom client_ip berisi IP klien yang valid (seperti 10.90.2.x atau 10.90.3.x), dan bukan IP proxy (10.90.1.4 atau 10.90.4.20).









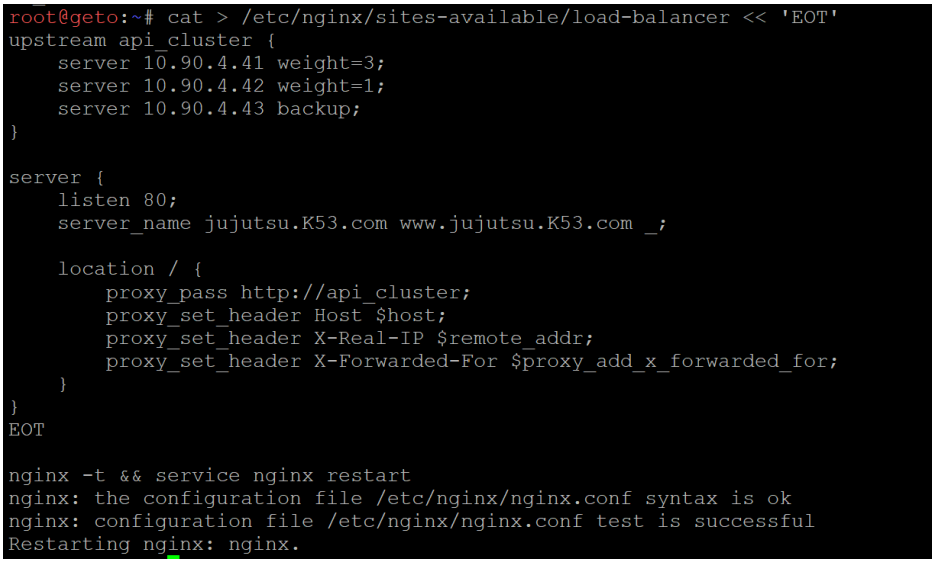
18.
jalanin digeto

```
cat > /etc/nginx/sites-available/load-balancer << 'EOT'
upstream api_cluster {
   server 10.90.4.41 weight=3;
   server 10.90.4.42 weight=1;
   server 10.90.4.43 backup;
}

server {
   listen 80;
   server_name jujutsu.K53.com www.jujutsu.K53.com _;

   location / {
      proxy_pass http://api_cluster;
      proxy_set_header Host $host;
      proxy_set_header X-Real-IP $remote_addr;
      proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
   }
}
EOT

nginx -t && service nginx restart
```



2. Skenario Uji Failover (Pembuktian kirara Mengambil Alih)

1. Matikan Nginx di nanami dan todo:
    - Di konsol nanami: service nginx stop
    - Di konsol todo: service nginx stop

```
root@nanami:~# service nginx stop
Stopping nginx: nginx.
```

```
root@todo:~# service nginx stop
Stopping nginx: nginx.
```

jalankan di itadori
curl -i http://jujutsu.K53.com/api/

```
itadori:~# curl -i http://jujutsu.K53.com/api/
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 14 Oct 2026 15:50:19 GMT
Content-Type: application/json
Transfer-Encoding: chunked
Connection: keep-alive

{"status":"success","worker":"kirara","client_ip":"10.90.2.10","db_log":"logged_to_yaga"}
```

Nyalakan Kembali nanami dan todo:

- Di konsol nanami: service nginx start
- Di konsol todo: service nginx start

Cek Log Akhir di Database (yaga & inumaki):
mysql -u root -D db_shibuya_K53 -e "SELECT * FROM request_logs ORDER BY id DESC LIMIT 5;"

```
root@yaga:~# mysql -u root -D db_shibuya_K53 -e "SELECT * FROM request_logs O
R BY id DESC LIMIT 5;"
+----+-------------+------------+---------------------+
| id | worker_name | client_ip  | created_at          |
+----+-------------+------------+---------------------+
| 22 | kirara      | 10.90.2.10 | 2026-10-14 15:50:19 |
| 21 | kirara      | 10.90.2.10 | 2026-10-14 15:49:06 |
| 20 | nanami      | 10.90.2.10 | 2026-10-14 15:39:48 |
| 19 | nanami      | 10.90.2.10 | 2026-10-14 15:37:34 |
| 18 | todo        | 10.90.2.10 | 2026-10-14 15:37:34 |
+----+-------------+------------+---------------------+
root@yaga:~#
```

```
root@inumaki:~# mysql -u root -D db_shibuya_K53 -e "SELECT * FROM request_logs O
RDER BY id DESC LIMIT 5;"
+----+-------------+------------+---------------------+
| id | worker_name | client_ip  | created_at          |
+----+-------------+------------+---------------------+
| 22 | kirara      | 10.90.2.10 | 2026-10-14 15:50:19 |
| 21 | kirara      | 10.90.2.10 | 2026-10-14 15:49:06 |
| 20 | nanami      | 10.90.2.10 | 2026-10-14 15:39:48 |
| 19 | nanami      | 10.90.2.10 | 2026-10-14 15:37:34 |
| 18 | todo        | 10.90.2.10 | 2026-10-14 15:37:34 |
+----+-------------+------------+---------------------+
root@inumaki:~#
```

jangan lupa nyalain lagi di nanami dan todo
service nginx start

cek apakah diitadori sudah normal
curl -s http://jujutsu.K53.com/api/

```
itadori:~# for i in {1..6}; do curl -s -o /dev/null -w "%{http_code} " http://jujutsu.K53.com/api/; done
200 200 200 200 200 200 itadori:~#
```

19.
jalankan digojo

```
cat > /root/gojo-cache-ratelimit.sh << 'EOF'
#!/bin/bash
export DEBIAN_FRONTEND=noninteractive

# 1. Buat direktori untuk menyimpan cache Nginx
mkdir -p /var/cache/nginx/jujutsu_cache

# 2. Definisikan Zone Rate Limiting (10 request per detik per IP, nama zona:
api_limit, memory: 10MB)
# Letakkan konfigurasi ini di luar blok server (di dalam blok http)
cat > /etc/nginx/conf.d/ratelimit.conf << 'EOT'
limit_req_zone $binary_remote_addr zone=api_limit:10m rate=10r/s;
EOT

# 3. Konfigurasi Utama Proxy Cache & Rate Limiting pada Gojo
cat > /etc/nginx/sites-available/gojo-proxy << 'EOT'
# Definisikan parameter cache proxy
proxy_cache_path /var/cache/nginx/jujutsu_cache levels=1:2
keys_zone=my_cache:10m max_size=10g inactive=60m use_temp_path=off;

upstream web_cluster {
    server 10.90.4.31;
    server 10.90.4.32;
}

server {
    listen 80;
    server_name jujutsu.K53.com www.jujutsu.K53.com _;

    # Arahkan trafik API ke Internal Load Balancer geto dengan Rate Limiting
    location /api/ {
        # Batasi 10 req/s, burst=20 agar request cepat tidak langsung error 503
        limit_req zone=api_limit burst=20 nodelay;

        proxy_pass http://10.90.4.20/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
```

```
    # Arahkan trafik web utama ke cluster web frontend dengan Proxy Caching (10
menit)
    location / {
        proxy_pass http://web_cluster;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;

        # Aktifkan cache
        proxy_cache my_cache;
        proxy_cache_valid 200 302 10m; # Cache valid 10 menit untuk HTTP 200 dan
302
        proxy_cache_valid 404 1m;

        # Tambahkan header X-Cache-Status
        add_header X-Cache-Status $upstream_cache_status;
    }
}
EOT

ln -sf /etc/nginx/sites-available/gojo-proxy /etc/nginx/sites-enabled/gojo-proxy
rm -f /etc/nginx/sites-enabled/default

nginx -t && service nginx restart
EOF

bash /root/gojo-cache-ratelimit.sh
```

```
root@gojo:~# cat > /root/gojo-cache-ratelimit.sh << 'EOF'
#!/bin/bash
export DEBIAN_FRONTEND=noninteractive

# 1. Buat direktori untuk menyimpan cache Nginx
mkdir -p /var/cache/nginx/jujutsu_cache

# 2. Definisikan Zone Rate Limiting (10 request per detik per IP, nama zona: api
_limit, memory: 10MB)
# Letakkan konfigurasi ini di luar blok server (di dalam blok http)
cat > /etc/nginx/conf.d/ratelimit.conf << 'EOT'
limit_req_zone $binary_remote_addr zone=api_limit:10m rate=10r/s;
EOT

# 3. Konfigurasi Utama Proxy Cache & Rate Limiting pada Gojo
cat > /etc/nginx/sites-available/gojo-proxy << 'EOT'
# Definisikan parameter cache proxy
proxy_cache_path /var/cache/nginx/jujutsu_cache levels=1:2 keys_zone=my_cache:10
m max_size=10g inactive=60m use_temp_path=off;

upstream web_cluster {
    server 10.90.4.31;
    server 10.90.4.32;
bash /root/gojo-cache-ratelimit.shault-proxy /etc/nginx/sites-enabled/gojo-proxy
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
Restarting nginx: nginx.
```

jalankan diitadori
curl -I http://10.90.1.4/

- Akses Pertama (Cache Belum Ada): Header akan menampilkan X-Cache-Status: MISS (artinya Nginx mengambil data baru dari backend maki/panda).
- Akses Kedua (Segera Sesudahnya): Header akan menampilkan X-Cache-Status: HIT (artinya Nginx menyajikan halaman langsung dari penyimpanan cache lokal tanpa membebani backend).

jalankan juga diitadori
for i in {1..25}; do curl -s -o /dev/null -w "%{http_code}\n" http://10.90.1.4/api/; done

Hasil yang Diharapkan:

- Beberapa request awal akan menghasilkan status 200 (sukses).
- Begitu melebihi batas (karena burst=20 terlampaui), request berikutnya akan langsung diblokir oleh Nginx dengan menghasilkan status 503 Service Temporarily Unavailable.

```
itadori:~# curl -I http://10.90.1.4/
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 14 Oct 2026 15:58:46 GMT
Content-Type: text/plain;charset=UTF-8
Connection: keep-alive
X-Cache-Status: MISS

itadori:~# curl -I http://10.90.1.4/
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 14 Oct 2026 15:59:47 GMT
Content-Type: text/plain;charset=UTF-8
Connection: keep-alive
X-Cache-Status: HIT

itadori:~# for i in {1..25}; do curl -s -o /dev/null -w "Request $i -> HTTP Status: %{http_code}\n" http://10.90.1.4/api/; done
Request 1 -> HTTP Status: 200
Request 2 -> HTTP Status: 200
Request 3 -> HTTP Status: 200
Request 4 -> HTTP Status: 200
Request 5 -> HTTP Status: 200
Request 6 -> HTTP Status: 200
Request 7 -> HTTP Status: 200
Request 8 -> HTTP Status: 200
Request 9 -> HTTP Status: 200
Request 10 -> HTTP Status: 200
Request 11 -> HTTP Status: 200
Request 12 -> HTTP Status: 200
Request 13 -> HTTP Status: 200
Request 14 -> HTTP Status: 200
Request 15 -> HTTP Status: 200
Request 16 -> HTTP Status: 200
Request 17 -> HTTP Status: 200
Request 18 -> HTTP Status: 200
Request 19 -> HTTP Status: 200
Request 20 -> HTTP Status: 200
Request 21 -> HTTP Status: 200
Request 22 -> HTTP Status: 200
Request 23 -> HTTP Status: 200
Request 24 -> HTTP Status: 503
Request 25 -> HTTP Status: 503
```

20.

jalankan di kusakabe
udhcpc -i eth0
echo "nameserver 10.90.1.2" > /etc/resolv.conf
apk update && apk add apache2-utils curl

- Benchmark halaman utama (dilayani dari proxy cache gojo)
jalankan di kusakabe
ab -n 600 -c 30 http://jujutsu.K53.com/

- Benchmark /api/ dengan rate limiting AKTIF
jalankan di kusakabe
ab -n 600 -c 30 http://jujutsu.K53.com/api/

- Benchmark /api/ baseline (rate limiting dinonaktifkan sementara)
jalankan di gojo
sed -i 's/limit_req zone/#limit_req zone/' /etc/nginx/sites-available/gojo-proxy
nginx -t && service nginx reload

jalankan di kusakabe
ab -n 600 -c 30 http://jujutsu.K53.com/api/

aktifkan kembali rate limiting
jalankan di gojo
sed -i 's/#limit_req zone/limit_req zone/' /etc/nginx/sites-available/gojo-proxy
nginx -t && service nginx reload
grep limit_req /etc/nginx/sites-available/gojo-proxy

- Pembuktian ukuran body (untuk menjelaskan "Failed requests" di ab)
jalankan di kusakabe
for i in 1 2 3 4 5 6 7 8; do curl -s http://jujutsu.K53.com/api/ | wc -c; done

- Analisis access log gojo
jalankan di gojo
awk '$1=="10.90.3.11"{print $7, $9}' /var/log/nginx/access.log | sort | uniq -c
grep -c "limiting requests" /var/log/nginx/error.log
tail -n 20 /var/log/nginx/access.log

HASIL BENCHMARK (dari kusakabe, ab -n 600 -c 30)

Metrik	/ (cache)	/api/ rate limit AKTIF	/api/ baseline (TANPA limit)
Waktu total	0,125 s	0,165 s	1,066 s

| Requests per second | 4786,86 | 3637,36 | 563,08 |
| Latensi mean (per request) | 6,27 ms | 8,25 ms | 53,28 ms |
|
Latensi p50 / p95 / p99	-	6 / 8 / 61 ms	58 / 78 / 83 ms
Latensi maksimum	-	77 ms	84 ms
Respons 2xx	600	22	600
Respons ditolak (Non-2xx, 503)	0	578	0
Failed requests (ab)	0	583 (semua Length)	149 (semua Length)
Connect / Receive / Exception	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0

ANALISIS

1. Halaman utama (/)
   RPS 4786,86 dan 0 request gagal. Performanya sangat tinggi karena gojo menyajikan halaman dari proxy cache (status HIT) sehingga server backend maki/panda tidak terbebani.

2. Baseline vs rate limiting aktif pada /api/
   - Baseline (tanpa limit) mencapai 563,08 req/s dengan latensi mean 53,28 ms. Semua 600 request sukses karena setiap request benar-benar diteruskan gojo -> geto -> worker API -> dicatat ke database yaga. Angka ini adalah kapasitas /api/ yang sebenarnya.
   - Saat rate limit aktif, RPS naik menjadi 3637,36 req/s dan latensi turun menjadi 8,25 ms. Ini BUKAN peningkatan kapasitas: gojo langsung membalas 503 untuk request yang melebihi kuota tanpa meneruskannya ke backend.
   - Hanya 22 request yang lolos (burst 20 + 1 request awal + sisa token selama uji berlangsung), sedangkan 578 request ditolak dengan 503.
   - Rate limiting terbukti melindungi backend dari lonjakan request.

3. Catatan tentang "Failed requests"
   Semua failed pada ab bertipe Length, bukan kegagalan sungguhan (Connect = 0, Receive = 0, Exceptions = 0).
   - Pada baseline, ab membandingkan panjang body dengan respons pertama (90 byte, dari nanami). Respons todo lebih pendek (88 byte) karena nama worker-nya lebih pendek. Hasil loop curl | wc -c menunjukkan urutan 90 90 88 90 90 90 88 90, yaitu 6 kali 90 (nanami) dan 2 kali 88 (todo), sesuai bobot load balancer 3:1. Karena itu 149 dari 600 request (~25%) dihitung "failed length".
   - Pada uji rate limit, body 503 (190 byte) berbeda dari body JSON (90 byte).
   - Jumlah request yang benar-benar ditolak mengikuti angka Non-2xx responses (578), bukan angka Failed.

ANALISIS ACCESS LOG GOJO (dari kusakabe 10.90.3.11)

Path	Status	Jumlah	Keterangan
/	200	600	Halaman utama, dilayani dari cache
/api/	200	622	22 (uji dengan limit) + 600 (baseline)
/api/	503	578	Ditolak rate limit

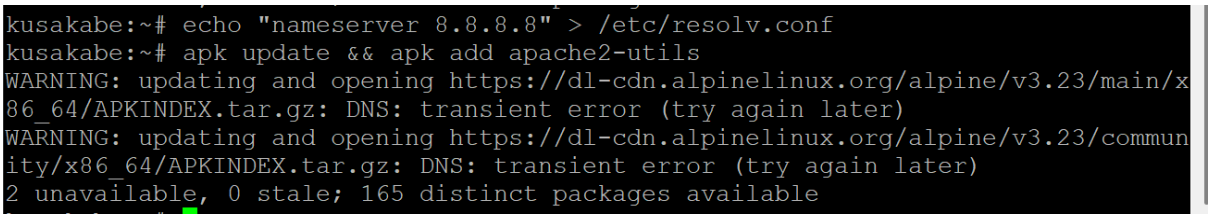
grep -c "limiting requests" pada error.log juga bernilai 578, sama dengan jumlah 503 di access log dan Non-2xx pada ab.

Pembedaan jenis kegagalan:

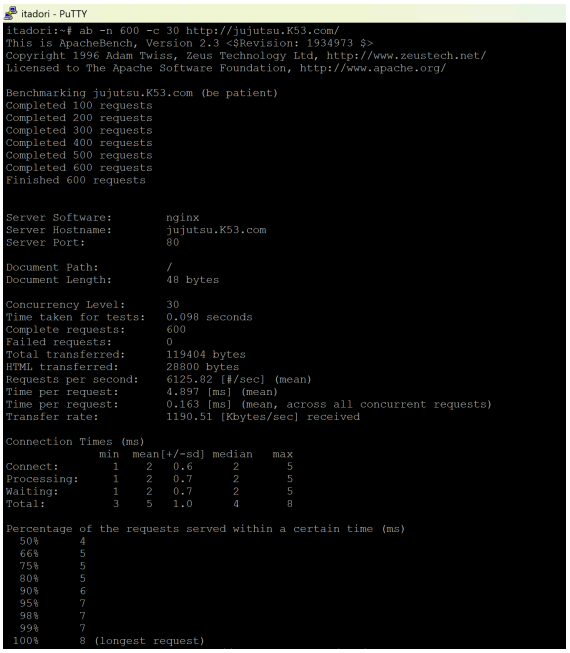
Jenis kegagalan	Jumlah	Bukti
Kegagalan aplikasi (502/504 dari upstream)	0	Tidak ada status 502/504 di access log
Kegagalan koneksi	0	ab: Connect = 0, Receive = 0, Exceptions = 0
Ditolak kebijakan rate limit	578	503 di access log + 578 entri "limiting requests"

KESIMPULAN

- Layanan web utama (/) stabil dan sangat cepat berkat proxy cache: 4786,86 req/s tanpa request gagal.
- Kapasitas /api/ tanpa pembatasan sekitar 563 req/s, latensi mean 53,28 ms, dan semua 600 request sukses.
- Rate limiting 10 r/s (burst 20) di gojo bekerja: dari 600 request beruntun hanya 22 diteruskan dan 578 ditolak dengan 503, sehingga backend terlindungi.
- Tidak ada kegagalan aplikasi maupun kegagalan koneksi. Seluruh respons non-2xx berasal dari kebijakan rate limit.



```
itadori:~# apk add apache2-utils
(1/4) Installing apr (1.7.6-r0)
(2/4) Installing libexpat (2.8.5-r0)
(3/4) Installing apr-util (1.6.4-r0)
(4/4) Installing apache2-utils (2.4.69-r0)
Executing busybox-1.37.0-r30.trigger
OK: 82.0 MiB in 169 packages
```



```
itadori:~# ab -n 600 -c 30 http://jujutsu.K53.com/api/
This is ApacheBench, Version 2.3 <$Revision: 1934973 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking jujutsu.K53.com (be patient)
Completed 100 requests
Completed 200 requests
Completed 300 requests
Completed 400 requests
Completed 500 requests
Completed 600 requests
Finished 600 requests


Server Software:        nginx
Server Hostname:        jujutsu.K53.com
Server Port:            80

Document Path:          /api/
Document Length:        88 bytes

Concurrency Level:      30
Time taken for tests:   0.152 seconds
Complete requests:      600
Failed requests:        594
   (Connect: 0, Receive: 0, Length: 594, Exceptions: 0)
Non-2xx responses:      578
Total transferred:      209842 bytes
HTML transferred:       111788 bytes
Requests per second:    3954.11 [#/sec] (mean)
Time per request:       7.587 [ms] (mean)
Time per request:       0.253 [ms] (mean, across all concurrent requests)
Transfer rate:          1350.48 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        0    3   0.7      3       5
Processing:     1    4   6.4      3      62
Waiting:        1    4   6.4      3      62
Total:          3    7   6.3      6      64

Percentage of the requests served within a certain time (ms)
  50%      6
  66%      6
  75%      6
  80%      6
  90%      7
  95%      8
  98%     27
  99%     46
 100%     64 (longest request)
```

```
kusakabe:~# udhcpc -i eth0
echo "nameserver 192.168.122.1" > /etc/resolv.conf
apk update && apk add apache2-utils
echo "nameserver 10.90.1.2" > /etc/resolv.conf
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 10.90.3.11, server 10.90.4.10
udhcpc: lease of 10.90.3.11 obtained from 10.90.4.10, lease time 1800
v3.23.6-59-g3461381db6a [https://dl-cdn.alpinelinux.org/alpine/v3.23/main]
v3.23.6-57-g67c83bfcab5 [https://dl-cdn.alpinelinux.org/alpine/v3.23/community]
OK: 27633 distinct packages available
(1/4) Installing apr (1.7.6-r0)
(2/4) Installing libexpat (2.8.5-r0)
(3/4) Installing apr-util (1.6.4-r0)
(4/4) Installing apache2-utils (2.4.69-r0)
Executing busybox-1.37.0-r30.trigger
OK: 82.0 MiB in 169 packages
```

kusakabe - PuTTY

```
kusakabe:~# ab -n 600 -c 30 http://jujutsu.K53.com/
This is ApacheBench, Version 2.3 <$Revision: 1934973 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking jujutsu.K53.com (be patient)
Completed 100 requests
Completed 200 requests
Completed 300 requests
Completed 400 requests
Completed 500 requests
Completed 600 requests
Finished 600 requests


Server Software:        nginx
Server Hostname:        jujutsu.K53.com
Server Port:            80

Document Path:          /
Document Length:        48 bytes

Concurrency Level:      30
Time taken for tests:   0.125 seconds
Complete requests:      600
Failed requests:        0
Total transferred:      119401 bytes
HTML transferred:       28800 bytes
Requests per second:    4786.86 [#/sec] (mean)
Time per request:       6.267 [ms] (mean)
Time per request:       0.209 [ms] (mean, across all concurrent requests)
Transfer rate:          930.27 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        1    3   0.7      3       5
Processing:     1    3   0.8      3       6
Waiting:        1    3   0.7      3       5
Total:          3    6   1.1      6      10

Percentage of the requests served within a certain time (ms)
  50%      6
  66%      6
  75%      6
  80%      7
  90%      7
  95%      8
  98%      9
  99%      9
 100%     10 (longest request)
```

```
kusakabe:~# ab -n 600 -c 30 http://jujutsu.K53.com/api/
This is ApacheBench, Version 2.3 <$Revision: 1934973 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking jujutsu.K53.com (be patient)
Completed 100 requests
Completed 200 requests
Completed 300 requests
Completed 400 requests
Completed 500 requests
Completed 600 requests
Finished 600 requests


Server Software:        nginx
Server Hostname:        jujutsu.K53.com
Server Port:            80

Document Path:          /api/
Document Length:        90 bytes

Concurrency Level:      30
Time taken for tests:   0.165 seconds
Complete requests:      600
Failed requests:        583
   (Connect: 0, Receive: 0, Length: 583, Exceptions: 0)
Non-2xx responses:      578
Total transferred:      209844 bytes
HTML transferred:       111790 bytes
Requests per second:    3637.36 [#/sec] (mean)
Time per request:       8.248 [ms] (mean)
Time per request:       0.275 [ms] (mean, across all concurrent requests)
Transfer rate:          1242.31 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        1    3   0.7      3       6
Processing:     1    5   8.0      3      74
Waiting:        1    4   8.0      3      74
Total:          3    7   7.9      6      77

Percentage of the requests served within a certain time (ms)
  50%      6
  66%      6
  75%      7
  80%      7
  90%      8
  95%      8
  98%     37
  99%     61
 100%     77 (longest request)
```

```
root@gojo:~# tail -n 20 /var/log/nginx/access.log
grep -c "limiting requests" /var/log/nginx/error.log
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
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10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
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acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
10.90.3.11 - - [15/Oct/2026:03:01:31 +0000] "GET /api/ HTTP/1.0" 503 190 "-" "Ap
acheBench/2.3"
578
```

```
kusakabe:~# apk add curl
OK: 82.0 MiB in 169 packages
kusakabe:~# curl -s http://jujutsu.K53.com/api/ | wc -c
88
kusakabe:~# curl -s http://jujutsu.K53.com/api/ | wc -c
90
kusakabe:~# for i in 1 2 3 4 5 6 7 8; do curl -s http://jujutsu.K53.com/api/ | tee /dev/stdout | wc -c; done
180
180
176
180
180
180
176
180
kusakabe:~# for i in 1 2 3 4 5 6 7 8; do curl -s http://jujutsu.K53.com/api/ | wc -c; done
90
90
88
90
90
90
88
90
```

```
root@gojo:~# grep limit_req /etc/nginx/sites-available/gojo-proxy
        limit_req zone=api_limit burst=20 nodelay;
```